W4 Face Recognition: Figures

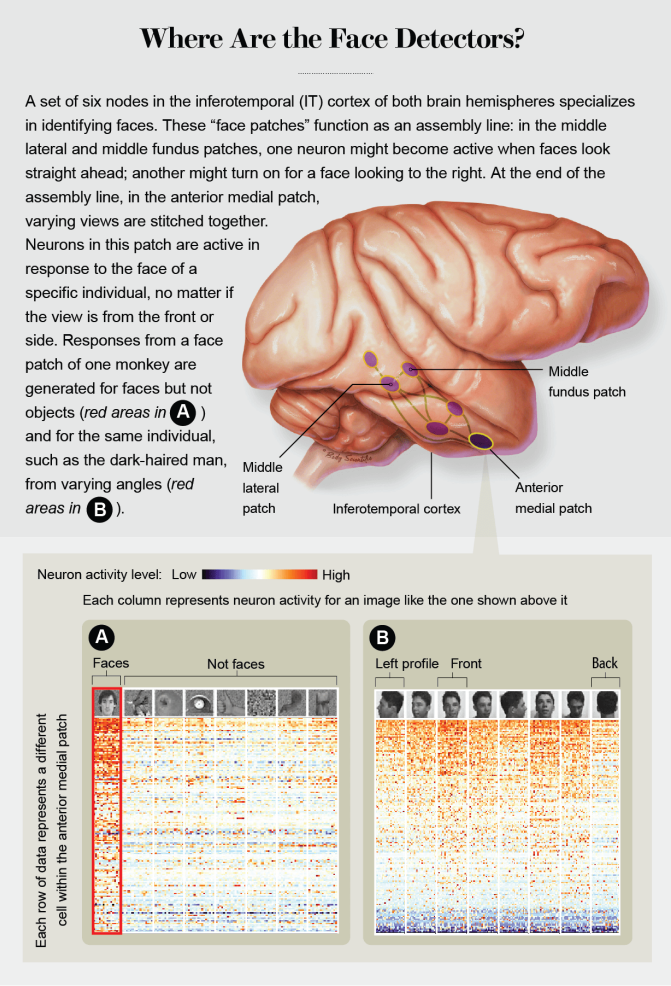
Thursday, October 15, 2026

READINGS Tsao, "Face Values" (2019); Chang & Tsao, "The Code for Facial Identity in the Primate Brain" (2017)

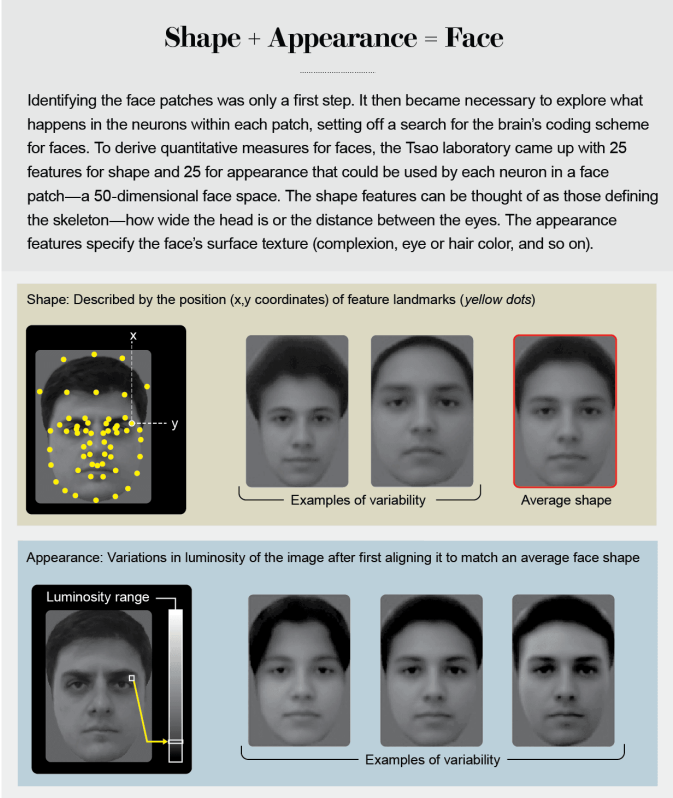
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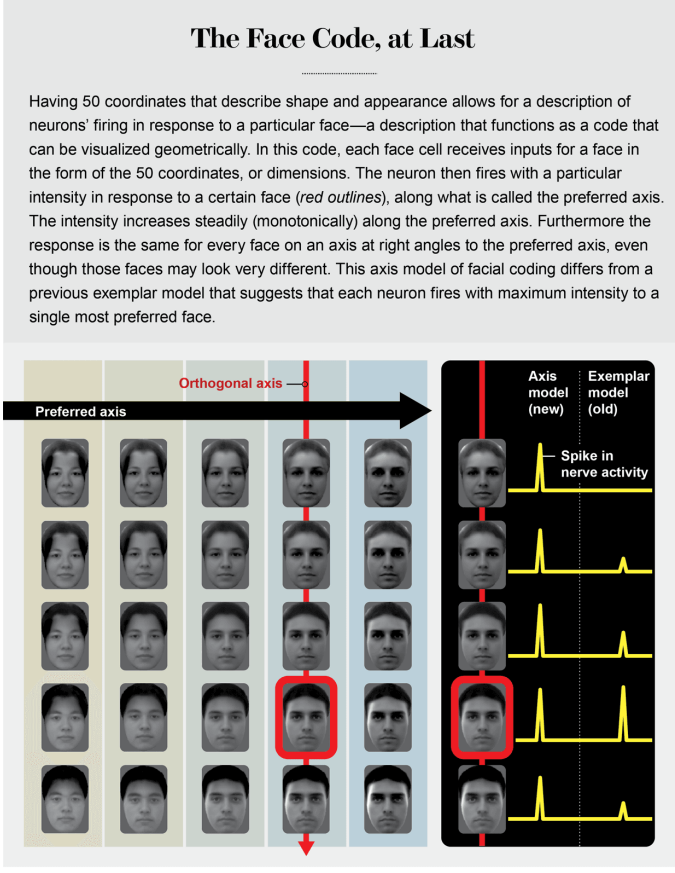
1 Tsao (2019), "Face Values"



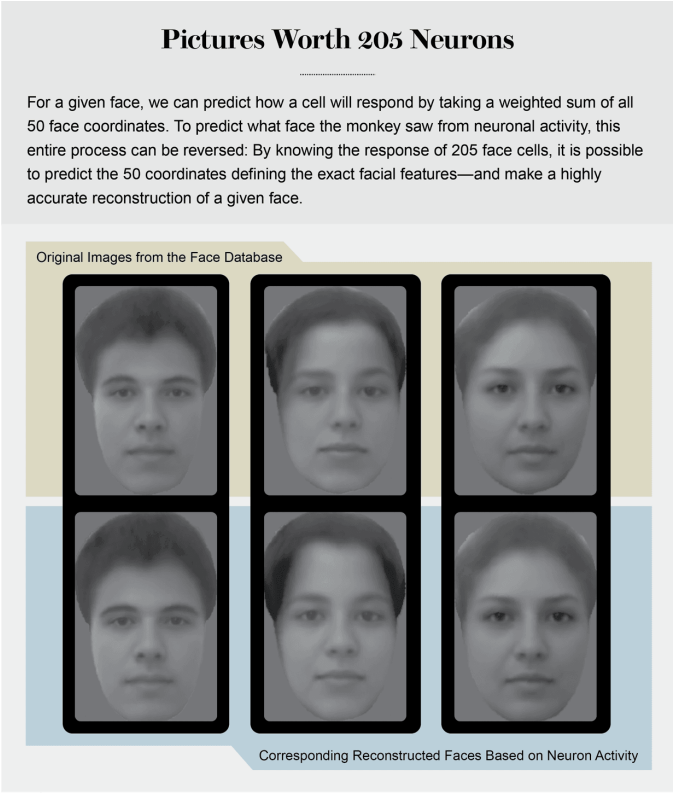
Credit: Body Scientific (*brain*); Source: "Functional Compartmentalization and Viewpoint Generalization within the Macaque Face-Processing System," by Winrich A. Freiwald and Doris Y. Tsao, in *Science*, Vol. 330; November 5, 2010 (*data grids and photo insets*) (p. 6).



Credit: Jen Christiansen (*graphic*); Source: "The Code for Facial Identity in the Primate Brain," by Le Chang and Doris Y. Tsao, in *Cell*, Vol. 169, No. 6; June 1, 2017 (*face images*) (p. 8).



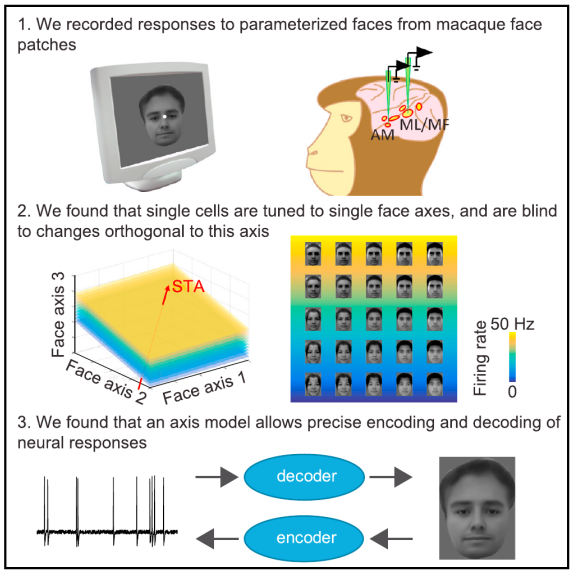
Credit: Jen Christiansen (*graphic*); Source: "The Code for Facial Identity in the Primate Brain," by Le Chang and Doris Y. Tsao, in *Cell*, Vol. 169, No. 6; June 1, 2017 (*face grid*) (p. 10).



Credit: Jen Christiansen (*graphic*); Doris Y. Tsao (*face images*) (p. 11).

2 Chang & Tsao (2017), "The Code for Facial Identity"

Graphical Abstract

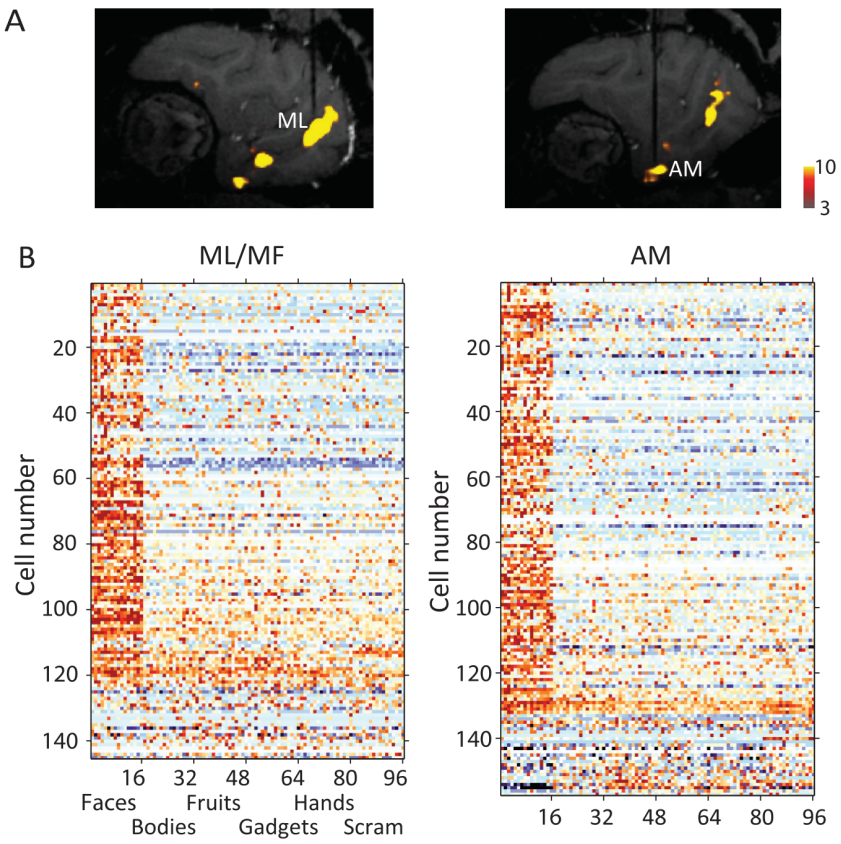


In Brief. Facial identity is encoded via a remarkably simple neural code that relies on the ability of neurons to distinguish facial features along specific axes in face space, disavowing the long-standing assumption that single face cells encode individual faces.

Highlights.

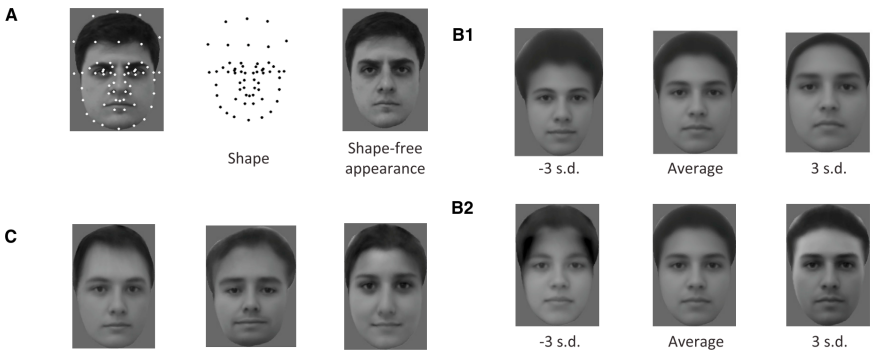
- Facial images can be linearly reconstructed using responses of ~200 face cells
- Face cells display flat tuning along dimensions orthogonal to the axis being coded
- The axis model is more efficient, robust, and flexible than the exemplar model
- Face patches ML/MF and AM carry complementary information about faces

Figure S1. Localization of Face Patches, Related to Figure 1

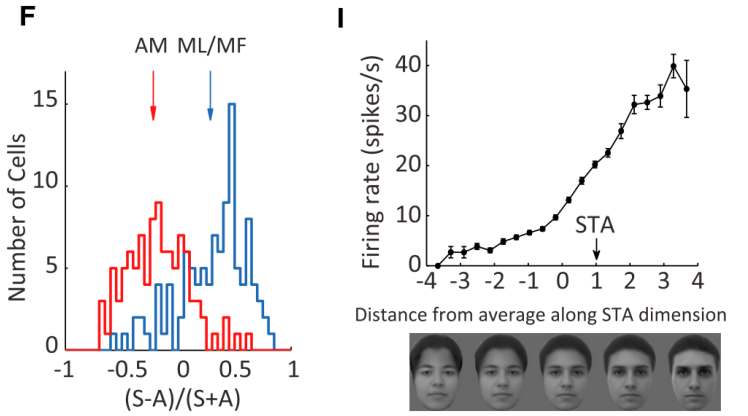


(A) Sagittal slices showing location of fMRI-identified face patches in one monkey targeted for recording; dark black line indicates electrode. **(B)** Neuronal responses (baseline-subtracted, averaged from 50 to 300 ms) to images of different categories recorded from the middle face patches (ML/MF, left) and the anterior face patch AM (right).

Figure 1. Complementary Representation of Facial Features by AM and ML/MF Populations

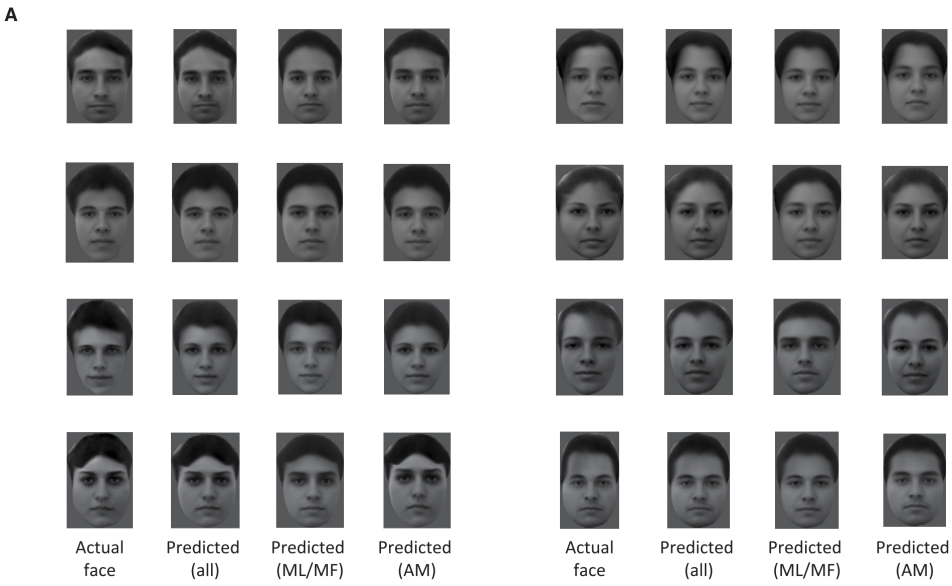


(A–C) Generation of parameterized face stimuli. (A) 58 landmark points were labeled on 200 facial images from a face database (FEI face database; example image shown on left). The positions of these landmarks carry shape information about each facial image (middle). The landmarks were smoothly morphed to match the average landmark positions in the 200 faces, generating an image carrying shape-free appearance information about each face (right). **(B) PCA was performed to extract the feature dimensions that account for the largest variability in the database.** The first principal component for shape (B1) and appearance (B2) are shown. **(C)** Example face stimuli generated by randomly drawing from a face space constructed by the first 25 shape PCs and first 25 appearance PCs.



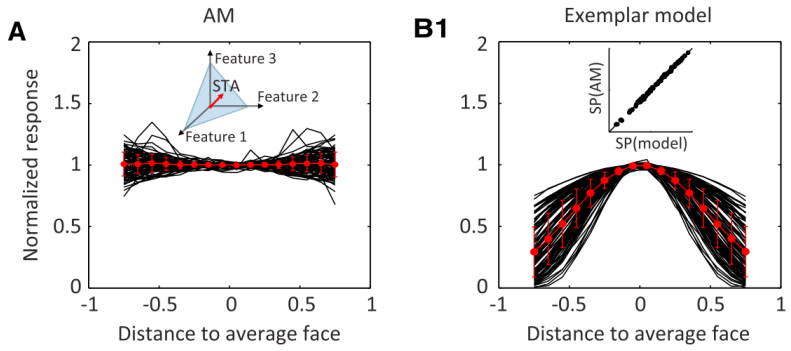
(F) Distribution of shape preference indices, quantified as the contrast between the vector length of shape and appearance STA for ML/MF and AM cells. Arrows indicate the average of each population ($p = 10^{-25}$, Student's t test). **(I) Response of a neuron in AM is plotted against distance between the stimulus and the average face along the STA axis.** Error bars represent SE. See also Figures S1 and S2.

Figure 3. Reconstruction of Facial Images Using Linear Regression



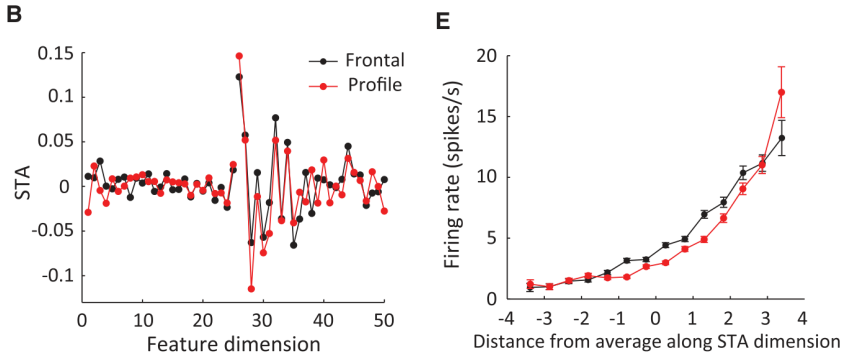
(A) Using facial features decoded by linear regression in Figure 2, facial images could be reconstructed. Predicted faces by three neuronal populations and the corresponding actual stimuli presented in the experiment are shown.

Figure 4. AM Neurons Display Almost Flat Tuning along Axes Orthogonal to the STA in Face Space



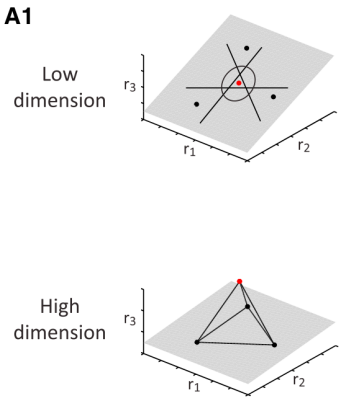
(A) For each neuron in AM, the STA was first computed, and then 2,000 random axes were selected and orthogonalized to the STA in the 25-d space of appearance features. Tuning functions along 300 axes accounting for the largest variability in the stimuli were averaged and fitted with a Gaussian function ($a \cdot e^{-(x^2/\sigma^2)} + c$). The center of the fit ($a + c$) was used to normalize the average tuning function. Red dots and error bars represent mean and SD of the population. **(B1)** Each simulated cell corresponds to one of the 200 real faces projected onto the 25-d face space of appearance features (exemplar face), and its response to an arbitrary face is a decreasing linear function of the Euclidean distance between the arbitrary face and the exemplar in the 25-d feature space. See also Figures S3, S4, S5, and S7.

Figure 6. The Axis Coding Model Is View Tolerant



(B) Spike-triggered average computed using 2,000 frontal stimuli (black) and 2,000 profile stimuli (red) for an example cell in AM. **(E)** Response of the neuron in (B) is plotted against distances between the stimuli and the average face along the STA axis for frontal and profile stimuli. The distance was rescaled so that STA corresponds to 1. Error bars represent SE.

Figure 7. An Axis-Metric Representation Is More Flexible, Efficient, and Robust for Face Identification



(A) An axis metric can perform as well as a distance metric on an identification task for high-dimensional representation but not for low-dimensional representation. (A1) For an identification task, a linear classifier is usually non-optimal for a low-dimensional space (upper); e.g., it is impossible to linearly separate the red dot from the black dots in the same plane, while a circular decision boundary defined by the distance to the red dot could easily perform the task. However, if the representation of these dots were high dimensional, it is much easier to separate the dots (lower).



(D) The linear relationship between neuronal responses and facial features ensures diverse tasks can be performed. The gray disks indicate face space. See also Figure S6.

3 Glossary (AI Generated)

Active appearance model A method from computer vision for turning a face into a list of numbers. Each face gets two descriptions: its shape (where the landmarks sit) and its appearance (the surface of the face once it has been warped to an average shape). Chang and Tsao used it to build their face space. (Fig. 1A–C)

AL The anterior lateral patch, in the middle of the chain, between ML/MF and AM. Its cells are less tied to one view: some respond to faces looking up, down or straight ahead, others to faces turned either left or right.

AM The anterior medial patch, the face patch farthest forward in the temporal lobe and the last stop in the chain. Its cells respond to who a face is, whatever the viewing angle, and they lean toward appearance features. Chang and Tsao recorded 99 cells here. (Fig. S1)

Appearance features The half of face space that describes the surface of a face rather than its geometry: skin tone, shading, the color of eyes and hair. They are measured after every face has been warped to the same average shape, so that differences in shape don't leak into them. (Fig. 1B)

Average face The face at the center of face space, where every feature takes its average value. Each axis measures how far a face departs from it in one particular way.

Axis One direction of variation in face space. Moving a face along an axis changes it in one coordinated way, like turning a single dial. Compare color: any color can be specified by how much red, green and blue it contains. Face space works the same way, with 50 dials instead of three.

Axis metric vs. distance metric Two ways a cell could measure a face. An axis metric asks how far along a certain direction the face lies, as you might ask how far north a city is. A distance metric asks how close the face is to one favorite face, as you might ask how far a city is from Chicago. Chang and Tsao argue that the brain uses the axis metric and that it is the better design. (Fig. 7)

Axis model The paper's main proposal. Each face cell takes the position of the incoming face in face space and reports how far it lies along one preferred axis. Its firing changes steadily as a face moves along that axis and not at all as the face moves at right angles to it. (Fig. 4A; Tsao, "The Face Code, at Last")

Decoding accuracy A test of how good a reconstruction is. The face decoded from the cells is compared with a lineup of real faces, and it counts as correct if its closest match is the face that was actually shown. With 40 faces in the lineup, the cells picked the right one 75% of the time; guessing would get 2.5%.

Decoding, encoding Encoding runs from the world to the brain: which pattern of firing a given face produces. Decoding runs the other way: reading a pattern of firing and working out which face caused it. The paper does both with the same simple linear rule. (Graphical Abstract)

Distance metric *see* axis metric vs. distance metric

Dot product A way of combining a face's coordinates with a cell's preferred axis: multiply each coordinate by the matching weight and add up the results. Geometrically, it measures how far the face's "shadow" falls along the axis, the way the sun casts a flagpole's shadow along the ground. It is the core operation of the axis model. Also called a projection.

Eigenfaces An older way of describing faces with numbers, popular in early computer face recognition: principal components analysis applied directly to the pixels of face photos, without first lining up the landmarks. The resulting dimensions look like blurry ghost faces, and they account for the cells' responses much less well than the shape and appearance axes do.

Encoding *see* decoding, encoding

Exemplar model The rival view. Each cell stores one particular face (an exemplar) and fires most for faces close to it, less and less as faces become more different. Think of each cell as a "wanted" poster for one face. Chang and Tsao's data favor the axis model instead. (Fig. 4B1)

Explained variance A score for how well a prediction tracks the data. If the predictions capture all the ups and downs of what is being predicted, the score is 100%. If they do no better than always guessing the average, it is 0%.

Face cell A neuron that fires much more to faces than to anything else. In a face patch, nearly every cell is a face cell. (Fig. S1B)

Face metamers Faces that look very different to us but make a particular cell fire at exactly the same rate, because they lie in that cell's null space. The term comes from color vision, where physically different lights that look like the same color are called metamers. Chang and Tsao built such faces for each cell they recorded. (Fig. 4A; Tsao, "The Face Code, at Last")

Face patch A small region of IT cortex packed with face cells. Macaques have six in each hemisphere, wired to one another and arranged from back to front like stations on an assembly line. Humans have a similar set. (Tsao, "Where Are the Face Detectors?"; Fig. S1A)

Face space An imagined space in which each point is a face. Each of its 50 dimensions is one way faces vary (25 for shape, 25 for appearance), and faces that look alike sit close together. Just as two coordinates fix a point on a map, 50 numbers fix a face here. (Tsao, "Shape + Appearance = Face")

Firing rate Neurons communicate with brief electrical pulses called spikes. The number of spikes a neuron fires per second is the usual measure of how strongly it is responding.

fMRI Functional magnetic resonance imaging: brain scanning that tracks changes in blood flow, which rises where neurons are active. It shows which regions light up, but not what individual cells are doing. Tsao used it to find the face patches before recording from single cells inside them.

Grandmother cell A hypothetical neuron that fires for one specific person and no one else, from any angle and in any light. Cells like this were reported in the human medial temporal lobe in 2005, one of them responding to pictures of Jennifer Aniston, hence the nickname "Jennifer Aniston cell." Tsao argues that the face patches don't work this way, though such cells may exist further on, in memory areas.

Holistic dimensions Most axes of face space don't change one isolated part, such as the length of the nose. Each changes several parts at once: the first shape axis, for example, moves the hairline, the width of the face and the height of the eyes together.

Invariance problem The challenge of recognizing the same thing despite changes in how it looks: a new angle, different lighting, greater distance, or something partly in front of it. The face patches seem to solve it for faces step by step, ending with AM's cells, which respond to the same person from any angle.

IT cortex Inferotemporal cortex, on the underside of the temporal lobe, roughly behind the temple. It is the last stage of the brain's visual pathway for recognizing objects, and it is where the face patches are.

Jennifer Aniston cell *see* grandmother cell

Landmarks Points marked by hand on key features of a face photo: the corners of the eyes, the tip of the nose, the edges of the mouth, the outline of the jaw. Chang and Tsao used 58 per face, and their positions define the face's shape. (Fig. 1A)

Leave-one-out A fair way to test a prediction method. Learn from all the data except one item, predict that item, and repeat for every item in turn. This guarantees the method is predicting faces it hasn't seen, not just remembering them.

Linear Describes a relationship in which effects simply add up and scale in proportion: double an input and you double its contribution. A weighted sum is linear; thresholds, ceilings and peaks are not. The surprise of the paper is how close to linear the face cells' computation turns out to be.

Linear regression The standard statistical method for finding the weighted sum of some quantities that best predicts another quantity. Chang and Tsao used it to learn how to turn the cells' firing rates into the 50 numbers that describe a face. (Fig. 3A)

Macaque The species of monkey studied in these experiments. Its visual system is organized much like ours.

Metamer *see* face metamers

Microstimulation Passing a tiny electrical current through an electrode to activate the neurons near its tip. Tsao's group used it to map which patches are connected to which, and to show that stimulating a face patch changes how a monkey sees faces.

ML/MF The middle lateral and middle fundus patches: two neighboring face patches in the middle of the temporal lobe, early in the chain. Their cells respond to particular views, such as faces turned to the left, and lean toward shape features. Chang and Tsao recorded 106 cells here. (Figs. S1, 1F)

Null space For a given cell, the set of changes to a face that the cell ignores: every change at right angles to its preferred axis. All the faces that differ only in these ways make the cell fire at the same rate. In a 50-dimensional face space this set is huge, with 49 of the 50 dimensions. (Tsao, "The Face Code, at Last")

Orthogonal axis An axis at right angles to a cell's preferred axis. Moving a face along it leaves the face's position on the preferred axis unchanged, and AM cells' firing barely changed along such axes. (Fig. 4A)

PCA *see* principal components analysis

Population code Information carried by the pattern of activity across many cells rather than by any single cell. In the axis model, each cell reports roughly one measurement of a face, and only the whole population pins down whose face it is, much as a single coordinate doesn't fix a place on a map. (Tsao, "Pictures Worth 205 Neurons")

Preferred axis The direction in face space that a cell cares about, estimated by its spike-triggered average. Moving a face one way along this axis makes the cell fire more, and moving it the other way makes it fire less. Also called the STA axis. (Figs. 1I, 6B)

Principal components analysis (PCA) A statistical technique that finds the directions in which a set of data varies most, in order of importance. Describing each item by its position along the top few directions captures most of the differences with only a few numbers. For people's heights and weights, for example, the first direction would be roughly overall body size. Chang and Tsao kept the top 25 directions for shape and the top 25 for appearance. (Fig. 1B)

Projection *see* dot product

Ramp-shaped tuning A response that rises steadily as a face moves along a cell's preferred axis, from low at one end to high at the other, with no peak at any particular face. It is the signature of the axis model. (Fig. 1I)

Shape features The half of face space that describes geometry, as given by where the landmarks sit: how wide the head is, how far apart the eyes are. (Fig. 1B)

Shape-free appearance What remains of a face photo after it has been warped so that its landmarks match the average shape: the skin, shading and coloring, with the geometry factored out. (Fig. 1A)

Single-unit recording Recording the spikes of individual neurons with a fine electrode inserted into the brain. Unlike fMRI, it shows exactly what single cells respond to.

Sparse cell A cell that responds strongly to only a few faces. Sparse cells in AM looked like grandmother cells, but Chang and Tsao found that they are axis cells with a steep threshold: they fire only for faces far out along their preferred axis, whoever those faces belong to.

Spike *see* firing rate

Spike-triggered average (STA) A way to find out what a cell "likes." Average all the faces shown to it, counting each face as many times as the spikes it produced. The result is a direction in face space, the cell's preferred axis, and it can be drawn as a face. (Fig. 6B)

STA *see* spike-triggered average

STA axis *see* preferred axis

Tuning How a cell's response changes as the stimulus changes. A cell is tuned to a feature if its firing goes up or down as that feature changes, much as a radio is tuned to pick up some stations and not others.

View-specific, view-invariant A view-specific cell responds to faces seen from one angle, such as straight on or in profile. A view-invariant cell responds to the same person from any angle. The face patches move from view-specific cells in ML/MF to view-invariant cells in AM. (Fig. 6)

Weighted sum Multiply each number by a weight that reflects its importance, then add them up. A course grade in which the final counts twice as much as each quiz is a weighted sum. In the axis model, a cell's firing is a weighted sum of a face's 50 features.